



Environmental, reproductive, and nutrient drivers of three-dimensional growth in the forest lichen *Lobaria pulmonaria*

Yngvar Gauslaa¹

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Abstract

Measuring growth dynamics has been fundamental to understanding lichen biology and ecology. In this integrative analysis, I synthesize and reanalyze published data on three-dimensional growth in the foliose cephalolichen *Lobaria pulmonaria* transplanted in pristine North American forests, encompassing thallus area expansion, biomass accumulation, thallus thickening, and reproductive investment across multiple environmental gradients. Best subset regression selection and linear mixed models were applied to identify the principal internal and external factors influencing growth, specifically thallus area expansion and thickening, quantified as increases in specific thallus mass. The analyses reveal pronounced intrathalline growth heterogeneity, significant trade-offs between reproductive investment and growth, and the multifactorial influences of bark pH, elevation, light regime, and phosphorus availability. Light was often a limiting growth factor within forest interiors, whereas excessive stand openness caused chlorophyll bleaching and reduced growth. Initial phosphorus enrichment enhanced thallus thickness but did not consistently promote area or biomass expansion. Taken together, these findings highlight the habitat specialization of *L. pulmonaria* and the importance of bark pH, light, hydration, and reproductive investment in shaping growth dynamics. The study advances the understanding of growth strategies in *L. pulmonaria* and provides a foundation for predicting responses to environmental change and guiding conservation efforts.

Keywords Epiphytic lichens · Intercalary growth · Relative growth rate · Thallus thickening · Hydration regime · Nutrient limitation

Introduction

Growth measurement has long been central to lichenological research, offering critical insights into lichen biology and ecology (e.g., Vallot 1896; Linkola 1918; Porter 1927; Beschel 1958; Armstrong and Bradwell 2011). Early studies focused primarily on apical and marginal expansion as the main modes of thallus growth (Hale 1970; Hill 1981). Later studies revealed that lichen growth is more complex, with thallus thickening making a substantial contribution to biomass accumulation (Porter 1927). For instance, Larson (1983) proposed that thallus area expansion in *Umbilicaria*

species is diffuse, while Valladares and Ascaso (1994) showed that their growth predominantly occurs in a broad zone surrounding the central umbilicus (see also Hestmark 1997). Additional structural evidence for diffuse growth came from observations of cell division in the cortices of umbilicate lichens (Sanders and de los Ríos 2017, 2018). Comparable morphological patterns indicated that diffuse growth may also occur in foliose lichens, such as *Lobaria pulmonaria* (Honegger 1993). In fruticose lichens, diffuse or intercalary growth has been documented in *Ramalina menziesii* (Sanders 1989), *R. usnea* (Sanders and Tokamov 2015), and *Usnea longissima* (Rolstad and Rolstad 2008). Lichen growth is therefore best viewed as species-specific and is currently classified into three main categories: (1) apical/marginal growth; (2) combined apical/marginal and intercalary growth; and (3) patchy intercalary growth (Honegger 1993, 2012).

Within foliose lichens, species of *Lobaria* are notable for their particularly rapid growth rates (Phillips 1969; Hale 1973). Building on this remarkable growth potential, the

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✉ Yngvar Gauslaa
yngvar.gauslaa@nmbu.no

¹ Faculty of Environmental Sciences and Natural Resource Management, Norwegian University of Life Sciences, P.O. Box 5003, Ås NO-1432, Norway

present study synthesizes and reanalyzes data from previous investigations to better understand growth patterns in *Lobaria pulmonaria*. It combines findings from studies conducted in pristine, mixed forests (Gauslaa and Goward 2012; Bidussi et al. 2013b), each covering distinct ecological gradients, and incorporates recent assessments of reproductive investment and spatial chlorophyll bleaching as an indicator of high-light damage (Gauslaa 2026). Because lichen growth involves changes in both thallus area, mass and consequently thallus thickness, the analysis focuses on data that capture three-dimensional growth increments. The overall aim is to identify the internal and external factors that most strongly shape growth in *L. pulmonaria*. Understanding these drivers is crucial not only for clarifying the species' ecology, but also for predicting its response to logging and environmental change and for guiding conservation efforts.

The study has three specific objectives: (1) to characterize the intrathalline pattern of area expansion within *L. pulmonaria*; (2) to test for trade-offs between reproduction and growth by examining relationships between thallus area and mass increments and recent measures of reproductive investment, with the hypothesis that marginal soralia, by disrupting marginal meristems, would be associated with stronger growth reductions than laminal soralia; and (3) to develop predictive models of thallus area growth and thickening using available lichen traits and environmental variables from studies in pristine North American forests (Gauslaa and Goward 2012; Bidussi et al. 2013b; Gauslaa 2026).

Materials and methods

Measurements of growth

Historically, lichen thallus growth has often been quantified as the percentage mass increase, linear extension of individual lobes, or diameter increment in circular thalli, as reviewed by Hale (1973) and Armstrong (1976). With the development of digital photography and computerized image analysis, measurements of thallus area increment have become increasingly common. However, area-based methods require standardization of hydration status at the beginning and end of study periods, because thallus area can shrink substantially during desiccation.

Relative growth rates, originally developed for plants (Evans, 1972), have since been adapted for lichens. Relative growth rate (RGR) expresses the increase in dry mass (DM) in $\text{mg g}^{-1} \text{ day}^{-1}$, while the relative thallus area growth rate (RT_AGR) quantifies the increase in thallus area (A) in $\text{mm}^2 \text{ cm}^{-2} \text{ day}^{-1}$. When time (t) is measured in days, these

metrics are calculated as: $\text{RGR} = (\ln(\text{DM}_{\text{end}}/\text{DM}_{\text{start}})) * 1000/\Delta t$, and $\text{RT}_A\text{GR} = (\ln(A_{\text{end}}/A_{\text{start}})) * 100/\Delta t$.

Lichen thickness typically varies substantially across a thallus. However, specific thallus mass, defined as DM per unit fully hydrated area ($\text{STM} = \text{DM}/A$), serves as a practical proxy for mean thallus thickness (Gauslaa and Coxson 2011). Growth in thallus thickness is expressed as: Percent change in STM = $(\text{STM}_{\text{end}} - \text{STM}_{\text{start}}) * 100/\text{STM}_{\text{start}}$.

Dataset compilation and experimental overview

A comprehensive dataset was assembled by integrating published 14-month growth data (RGR, RT_AGR , and percent change in STM) for *L. pulmonaria* thalli transplanted into pristine inland forests of British Columbia (Gauslaa and Goward 2012; $N=208$; Bidussi et al. 2013b; $N=94$), together with recent quantifications of soralia for all transplants included in those studies (Gauslaa 2026). One study (Gauslaa and Goward 2012) transplanted thalli onto pairs of neighboring, similar-sized spruce trees in dripzone and non-dripzone position, either beneath or outside *Populus* canopies, on a gently undulating valley floor. The other study (Bidussi et al. 2013b) used spruce trees distributed along an adjacent mountain slope ranging from 550 to 1650 m a.s.l. This expanded dataset enables more robust assessment of growth and facilitates analyses of how reproductive traits, such as soralia, influence lichen growth rates.

Both transplantation experiments used *L. pulmonaria* transplants collected simultaneously from a single humid, partially shaded source stand dominated by juvenile and early-reproducing thalli. These studies were conducted concurrently from May 2010 to August 2011, thereby eliminating confounding effects of study and year or differing source material. This is particularly important because lichen growth rates can vary substantially among years (e.g., Vallet 1896; Phillips 1969). The studies spanned environmental gradients in pristine forests that regenerated after a stand-replacing wildfire in 1926, including bark pH, elevation, and site factors reflecting canopy openness. Indirect site factor (ISF) and direct site factor (DSF) describe the proportion of diffuse and direct solar radiation, respectively, reaching a lichen transplant (Anderson 1964).

Transplants were randomly assigned between the two experiments and subjected to similar methodologies. As a phosphorus (P) pretreatment, 50% of all transplants, selected at random, were submerged for 5 min in a 500 mg L^{-1} K_2PO_4 , whereas the remaining 50% served as controls and were submerged in distilled water. This treatment added both P and K to the transplants, but Gauslaa and Goward (2020) found no growth-stimulating effect of K in the local canopy throughfall on *L. pulmonaria*, suggesting that K was not a limiting factor. On each tree, lichen transplants

were attached to two separate branches. Further details of the experimental design and study area are provided in the original publications (Gauslaa and Goward 2012; Bidussi et al. 2013b; Gauslaa 2026).

Statistical methods

Data from five thalli were excluded due to partial or complete loss. Growth rates could not be calculated for ten thalli (3.3%) because of loss, fragmentation, or outlying STM values attributed to measurement or labeling errors. Means are reported $\pm$ standard error.

Model selection was performed using best subset regressions in Minitab Statistical Software 22, with variables log-transformed where appropriate. For analyses of variation in RT_AGR , RGR, and percent change in STM, the following explanatory variables were initially included: initial thallus area (A_{Start} ; mean = $13.76 \pm 0.25 \text{ cm}^2$, range = 5.7–29.9 cm^2); initial dry mass (DM_{Start} ; mean = $118.1 \pm 2.5 \text{ mg}$, range = 40–321 mg); initial specific thallus mass (STM_{Start} ; mean = $8.5 \pm 0.5 \text{ mg cm}^{-2}$, range = 6.1–11.7 mg cm^{-2}); change in STM (mean = $17.5 \pm 0.5\%$, range = -2.5–44.5%); bark pH (mean = 4.88 ± 0.04 , range = 3.95–6.77); indirect site factor (ISF; mean = 0.286 ± 0.003 , range = 0.179–0.457); direct site factor (DSF; mean = 0.230 ± 0.006 , range = 0.041–0.467); elevation (mean = $829 \pm 16 \text{ m}$, range = 550–1650 m); number of soralia at the end of the experiment per cm^2 thallus area (used as proxy for reproductive investments; mean = 3.36 ± 0.15 , range = 0–13.6); number of laminal soralia at the end of the experiment per cm^2 thallus area (best correlated with % increase in STM; mean = 0.43 ± 0.05 , range = 0–6.3); and chlorophyll bleaching at the end of the experiment (mean = $3.2 \pm 0.5\%$, range = 0–70%).

After identifying the optimal subset of predictors, the resulting model was further evaluated using a linear mixed model in Jamovi 2.6.26 with tree ID (the recipient of transplants) included as a cluster variable to account for potential non-independence of samples within trees.

Results and discussion

Intrathalline variability in growth rates

A typical *L. pulmonaria* thallus (Fig. 1) exhibited both apical/marginal and intercalary growth as proposed by Honegger (1993). In the illustrated specimen, the percent increase in mass (104%) exceeded the percent increase in area (83.9%) over the 14-month transplantation period, resulting in an 11.1% increase in STM and thus in thallus thickness. Expressed as relative growth, the RGR was $1.63 \text{ mg g}^{-1} \text{ d}^{-1}$ and the corresponding RT_AGR was $0.139 \text{ mm}^2 \text{ cm}^{-2} \text{ d}^{-1}$.

Growth was highly heterogeneous within the thallus, with individual lobes expanding markedly at different rates (Fig. 1). This pattern aligns with earlier reports of decoupled growth rates among separate lobes within a thallus (Linkola 1918; Phillips 1969). In this individual, lobe width increased more than lobe length, and the greatest expansion occurred in submarginal depressions between reticulated ridges in rapidly growing lobes (Fig. 1).

Species of *Lobaria* are recognized for their higher growth rates relative to other foliose lichens (Phillips 1969; Hale 1973), even when growing alongside them in favorable habitats such as open forests (Gauslaa and Goward 2012; Bidussi et al. 2013b) and forest gaps (Bidussi and Gauslaa 2015). The combination of pronounced intercalary growth and strong intrathalline variation, as visualized in Fig. 1, likely contributes to the fast growth observed in *L. pulmonaria*.

Balancing growth and reproduction: patterns and trade-offs

Over the full 14-month observation period, the mean RT_AGR for all *L. pulmonaria* transplants was $0.077 \pm 0.002 \text{ mm}^2 \text{ cm}^{-2} \text{ d}^{-1}$ (range = -0.007–0.158 $\text{mm}^2 \text{ cm}^{-2} \text{ d}^{-1}$, $N=298$), corresponding to $40.5 \pm 1.2\%$ area growth (range = -3–100%), and the mean RGR was $1.13 \pm 0.02 \text{ mg g}^{-1} \text{ d}^{-1}$ (range = 0.14–1.88 $\text{mg g}^{-1} \text{ d}^{-1}$), equivalent to a $64.6 \pm 1.4\%$ increase in dry mass (range = 6–128%). Greater reproductive investment was associated with lower RT_AGR and RGR (Table 1). Thallus area expansion was slightly more negatively affected by reproductive traits than mass growth (Table 1). Marginal soralia were more strongly associated with reduced growth rates than laminal soralia (Table 1), possibly because their formation disrupts apical and marginal meristems. However, 99% of transplants still showed positive RT_AGR , indicating that marginal soralia did not fully prevent thallus area expansion. Overall, total (marginal + laminal soralia) reproductive investments showed stronger negative correlation coefficients with growth rates than either soralia category alone (Table 1). At the end of experiment, marginal soralia were 6.8 times more abundant and covered 11.6 times more thallus area than laminal soralia (Gauslaa 2026), making it difficult to attribute stronger growth-retarding effects solely to their marginal position. Thus, the hypothesis that marginal soralia exert a stronger inhibitory effect on growth than laminal soralia received limited support.

Recognition of a trade-off between reproduction and growth in lichens dates back to early studies (e.g., Beschel 1958). In the present study, the final total number of soralia per thallus area showed the strongest relationship with RT_AGR (Fig. 2), although the effect was weak despite being

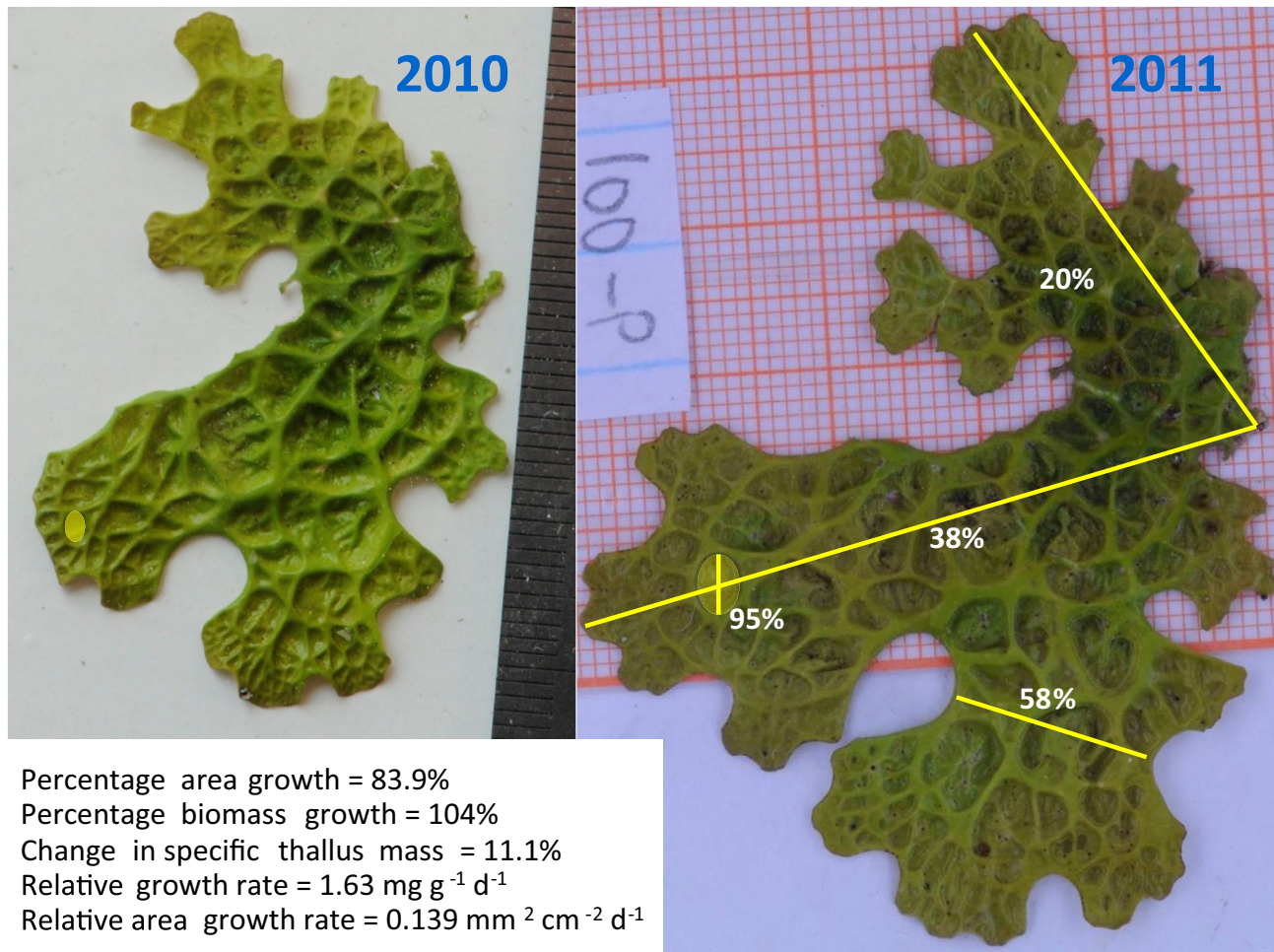


Fig. 1 A non-sorediate *Lobaria pulmonaria* dripzone transplant at 700 m receiving no P-treatment and with a bark pH of 6.4, shown at the same scale at the start (left) and end (right) of a 14-month field

transplantation. Percent linear growth measured along the marked lines is shown. The inset (lower left) summarizes various growth measures for the thallus after 14 months

highly significant ($R^2_{\text{adj}} = 8.74\%$; $P < 0.001$). A similar but slightly weaker trade-off was found for RGR ($R^2_{\text{adj}} = 7.31\%$; $P < 0.001$). A greater representation of fully mature thalli would likely have strengthened this relationship. Nevertheless, the results are consistent with previous findings from both boreal and Mediterranean climates (Gauslaa 2006; Merinero et al. 2015), echoing the classical growth-reproduction trade-off known from perennial plants (Harper 1977). This trade-off may not be obligatory in lichens, because soralia contain photobiont cells and may therefore contribute to the carbon balance. However, the extent of this contribution is uncertain, given the lower chlorophyll content in sorediate compared to non-sorediate lobes (Tretiach and Carpanelli 1992). The delayed onset of reproduction reported in *L. pulmonaria* (Scheidegger et al. 1997; Scheidegger and Goward 2002; Larsson and Gauslaa 2011) likely contributes to its relatively high growth rates compared to those of many other lichen species.

None of the reproductive parameters in Table 1 were significantly related to changes in thallus thickness ($P > 0.05$; data not shown). The strongest, though still weak, correlation was with laminal soralia at the end, expressed as percent of thallus area ($r = 0.101$; $P = 0.084$).

Multifactorial modelling of thallus area expansion and biomass change

The scatter in the regression between reproduction and growth in *L. pulmonaria* (Fig. 2) reflects the influence of environmental factors on growth rates (e.g., Gauslaa et al. 2007; Gauslaa and Goward 2012, 2020; Bidussi et al. 2013b). Best-subset multiple regression was therefore applied to the dataset spanning various environmental gradients and lichen traits and characterized by low multicollinearity (variance inflation factors: 1.00–1.37). The resulting best-subset model for $RT_{\text{A}}\text{GR}$ explained a substantial portion of variance ($R^2_{\text{adj}} = 58.9\%$; $P < 0.001$). Run as a linear mixed

Table 1 Pearson correlation coefficient with corresponding *P*–levels for relative growth rate (RGR) and relative thallus area growth rate (RT_AGR) versus (1) measured soralia parameters in 292–302 *Lobaria pulmonaria* thalli before (start) and after (end) a 14-month transplantation period in pristine mixed forests in British Columbia and (2) environmental factors experienced by the lichen transplants

	RGR	<i>P</i> -level	RT _A GR	<i>P</i> -level
At start:				
Laminal soralia, number thallus ⁻¹	-0.138	0.018	-0.192	0.001
Marginal soralia, number thallus ⁻¹	-0.233	<0.001	-0.261	<0.001
All soralia, number thallus ⁻¹	-0.235	<0.001	-0.270	<0.001
Soralia size, mm ⁻²	-0.149	0.010	-0.122	0.037
Marginal soralia, number cm ⁻²	-0.250	<0.001	-0.265	<0.001
Laminal soralia, number cm ⁻²	-0.134	0.021	-0.182	0.002
All soralia, number cm ⁻²	-0.252	<0.001	-0.274	<0.001
Marginal soralia, percent cover	-0.225	<0.001	-0.256	<0.001
Laminal soralia, percent cover	-0.137	0.018	-0.199	0.001
All soralia, percent cover	-0.228	<0.001	-0.265	<0.001
At end:				
Laminal soralia, number thallus ⁻¹	-0.061	0.300	-0.126	0.031
Marginal soralia, number thallus ⁻¹	-0.108	0.064	-0.101	0.085
All soralia, number thallus ⁻¹	-0.106	0.070	-0.125	0.033
Soralia size, mm ⁻²	0.000	0.999	-0.002	0.979
Marginal soralia, number cm ⁻²	-0.296	<0.001	-0.303	<0.001
Laminal soralia, number cm ⁻²	-0.123	0.036	-0.175	0.003
All soralia, number cm ⁻²	-0.276	<0.001	-0.301	<0.001
Marginal soralia, percent cover	-0.229	<0.001	-0.246	<0.001
Laminal soralia, percent cover	-0.149	0.011	-0.210	<0.001
All soralia, percent cover	-0.234	<0.001	-0.262	<0.001
Change in percent cover of all soralia	-0.171	<0.001	-0.199	<0.001
Environmental factors and change in STM:				
Elevation, m	-0.472	<0.001	-0.597	<0.001
Bark pH	0.463	<0.001	0.414	<0.001
Indirect site factor	0.196	<0.001	0.307	<0.001
Direct site factor	-0.104	0.077	0.044	0.454
Percent bleaching	-0.310	<0.001	-0.322	<0.001
Percent change in STM	0.237	<0.001	-0.257	<0.001

Growth data, elevation, bark pH, and site factors from Gauslaa and Goward (2012); Bidussi et al. (2013b); reproductive traits from Gauslaa (2026). RGR and RT_AGR were not related to transplant size. Significant correlations at *P*<0.05 are marked in **bold**. The correlation coefficient between RGR and RT_AGR was 0.878; *P*<0.001

model with transplant tree ID as the clustering variable, the fixed effect accounted for 61.1% of the variance (R^2_{marginal}), while inclusion of tree ID increased the explained variance to 68.2% ($R^2_{\text{conditional}}$; Table 2). The model identified two internal predictors (final number of soralia per thallus area and change in STM) and four external predictors (elevation, bark pH, chlorophyll bleaching due to excess light exposure, and ISF) as significant. High RT_AGR was associated with lower elevation, fewer soralia, less bleaching, smaller increases in STM, higher bark pH, and higher ISF. Five variables were highly significant (*P*<0.001), whereas ISF was only weakly significant (*P*=0.015; Table 2). Most significant factors in the RT_AGR model had similar predictive power (Table 2), underscoring the multifactorial nature of lichen growth.

Combining data from studies conducted along differing environmental gradients increased statistical power and

model robustness. Reproductive investment accounted for a significant, but not dominant, portion of explained variance. Incorporating the reproductive variable (Gauslaa 2026) improved the RT_AGR model's explanatory power and slightly increased the significance of the other included factors.

A 3-D plot (Fig. 3) illustrates the combined relationship of soralia abundance and bark pH with thallus area expansion: thalli with many soralia on acidic bark exhibited the slowest growth, while those with few soralia on bark with higher pH grew the fastest. Percentage bleaching, resulting from excessive light (Gauslaa et al. 2019), was another important predictor (Table 2), reflecting damage that reduced growth in shade-adapted source transplants transplanted to brighter sites (e.g., Gauslaa and Solhaug 2000).

Notably, 7.1% of variance in the RT_AGR model was attributable to the random effect of tree identity, while

Fig. 2 The relationship between relative thallus area growth and reproductive efforts in transplants of *Lobaria pulmonaria*. $RT_A GR = 0.0896 - 0.00383 \times \text{soralia number per } A_{\text{end}}$; $R^2_{\text{adj}} = 8.74\%$; $P < 0.001$. Data were combined from Gauslaa and Goward (2012); Bidussi et al. (2013b); Gauslaa (2026)

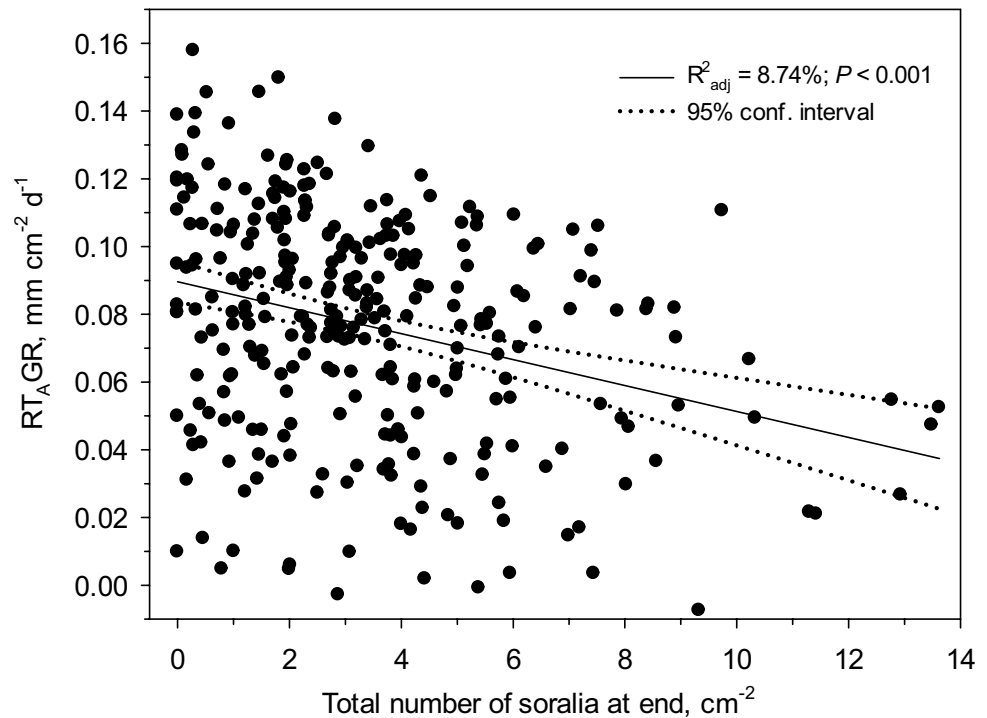


Table 2 The best subset multiple regression model for relative thallus area growth ($\text{mm}^2 \text{cm}^{-2} \cdot \text{day}^{-1}$) in *Lobaria pulmonaria* run as linear mixed model after adding the cluster variable: “tree ID” ($R^2_{\text{marginal}} = 61.1\%$; $R^2_{\text{conditional}} = 68.2\%$)

Term	Coeff $\pm$ SE ($\times 10^{-3}$)	t-value	P-value
Intercept	75.9 $\pm$ 177	42.87	<0.001
Elevation, m	-0.053 $\pm$ 0.006	-9.22	<0.001
Soralia _{End} , number cm^{-2}	-3.920 $\pm$ 0.469	-8.36	<0.001
Bleaching _{End} , % of area	-1.000 $\pm$ 0.165	-6.04	<0.001
Bark pH	13.76 $\pm$ 2.72	5.06	<0.001
Change in STM, %	-0.770 $\pm$ 0.155	-4.97	<0.001
Indirect site factor	76.6 $\pm$ 31.1	2.46	0.015

Relative thallus area growth ($RT_A GR$), elevation, bark pH, specific thallus mass at the end (STM_{End}), and the indirect site factors were retrieved from Gauslaa and Goward (2012); Bidussi et al. (2013b), while Soralia_{End} (which was the soralia variable with the best predictive power for $RT_A GR$) and percent bleached area of the thallus at the end (Bleaching_{End}) were quantified by Gauslaa (2026). Log-transformations of variables resulted in weaker models

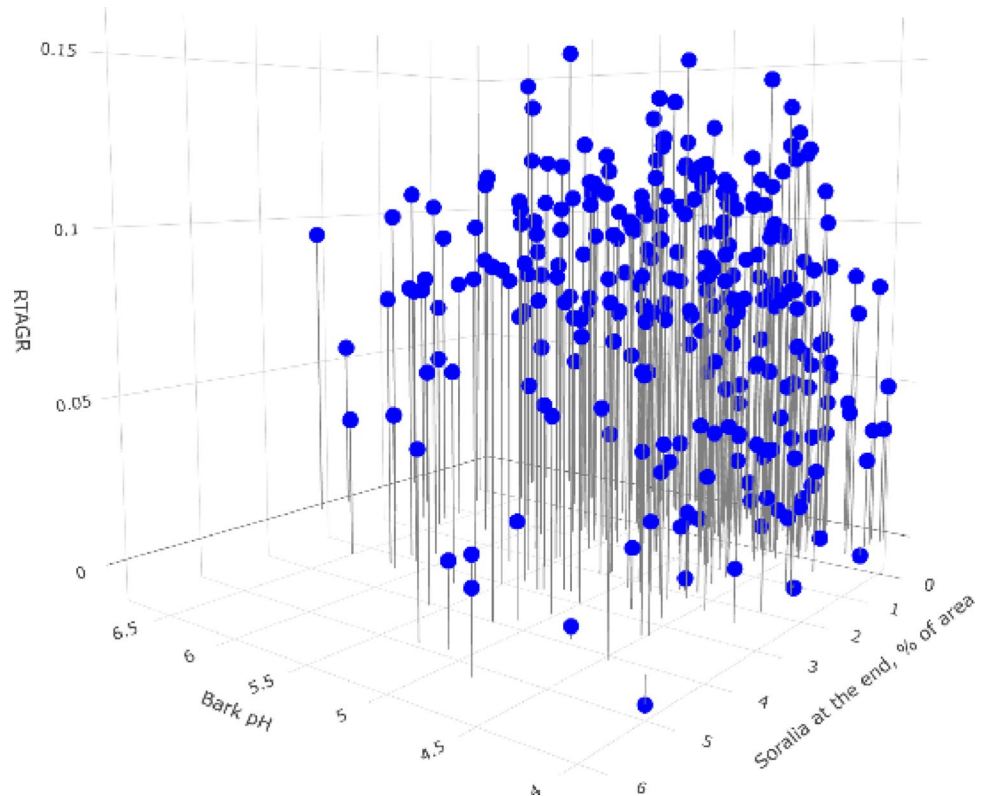
31.8% of variance remained unexplained (Table 2). In moist climates, light is often the most important driver of lichen growth (Porter 1927). In this relatively dry valley, however, (636–687 mm rainfall on the valley floor, increasing with elevation; Wang et al. 2012), ISF was the least significant factor, likely because all transplant trees were in relatively open stands with sufficient light. Indirect light appears more important for thallus area expansion than direct sunlight, which causes rapid desiccation and physiological inactivity. Open-shade habitats, rich in diffuse skylight but low in direct sunlight (Stoutjesdijk and Barkman 1987), provide optimal conditions for many lichens (Palmqvist and Sundberg 2000;

Palmqvist et al. 2008; Borge and Ellis 2024) while excess direct light can damage *L. pulmonaria* (Gauslaa and Solhaug 1996, 2000).

Site-specific water availability was not measured. Although rainfall is relatively consistent across the study area, variation in rainfall strongly influences lichen growth rates at broader spatial scales (Kärenlampi 1970; Gauslaa et al. 2007). At local scales, tree architecture redistributes water within the canopy. Taller trees channel throughfall more efficiently away from lichen sites on lower branches than smaller trees, as observed during rainy days (Gauslaa and Goward 2012). At the same time, tall canopies reduce nocturnal reradiation from lower branches, thereby diminishing the nocturnal cooling needed for dew formation. Air humidity and dewfall, both of which are at least as important for lichens as rainfall (Gauslaa 2014), vary strongly at small spatial scales (Bidussi et al. 2013b). Nocturnal dewfall promotes lichen growth by ensuring full turgor pressure and facilitating thallus area expansion (Bidussi et al. 2013a). Where assessed, the height of trees used for transplantation negatively affected growth of *L. pulmonaria* (Gauslaa and Goward 2012, 2020). Tree height and canopy density likely account for much of the variance associated with the random tree effect through their influence on hydration. Including direct measures of hydration sources would probably further improve future growth models.

Acclimatization to new environments may also reduce lichen growth rates (Gauslaa and Goward 2020; Kann et al. 2025). Most transplants experienced a shift from shaded to more illuminated transplantation sites, potentially triggering

Fig. 3 Relative thallus area growth (RT_{AGR}) of *Lobaria pulmonaria* transplants in relation to bark pH and percent cover of soralia grown in pristine forests for 14 months (multiple regression: $R^2_{adj}=24.1\%$; $P<0.001$). The simple regression between RT_{AGR} and percent cover of soralia at the end had an $R^2_{adj}=6.6\%$; $P<0.001$, whereas the simple regression between RT_{AGR} and bark pH had $R^2_{adj}=16.8\%$; $P<0.001$. Data were combined from Gauslaa and Goward (2012); Bidussi et al. 2013b); Gauslaa (2026)



growth-retarding acclimatization responses. Furthermore, subsequent studies including a larger range of forest habitats showed that canopy throughfall chemistry (high Ca, P and low Zn, S) was more strongly associated with high growth rates of *L. pulmonaria* than bark pH alone (Gauslaa and Goward 2020). More detailed chemical assessments could therefore improve growth predictions. Biological interactions, such as grazing by invertebrates, may also influence growth (Asplund and Gauslaa 2008; Gauslaa et al. 2018; Haughian et al. 2025), although no grazing marks were observed here.

Applying the same predictors in a linear mixed model for RGR yielded slightly weaker results ($R^2_{marginal}=59.5\%$; $R^2_{conditional}=66.8\%$; model not shown), although all variables remained significant. Given the strong correlation between RT_{AGR} and RGR ($r=0.878$), this similarity is not surprising. Notably, unlike RT_{AGR} , RGR was positively associated with changes in STM (Table 1).

Overall, the significant influence of multiple environmental factors on growth rates—and thus fitness—of *L. pulmonaria* supports its status as a habitat specialist (e.g., Otálora et al. 2011).

Determinants and patterns of thallus thickening

Both increased lobe thickness (Gauslaa and Coxson 2011) and a deeper reticulate morphology (Phinney 2019) increase

STM in *L. pulmonaria*. STM is a flexible functional trait that shapes the ecologically important internal water-holding capacity ($WHC_{internal}$), expressed either as mm water or as water mass per unit thallus area, across macrolichen growth forms (Gauslaa 2014; Esseen et al. 2015; Phinney et al. 2022). Higher STM prolongs active hydration periods in lichens (Gauslaa and Solhaug 1998; Gauslaa et al. 2017; Ås Hovind et al. 2020), while a more pronounced reticulate pattern forms external pockets that retain rainwater and thereby extend hydration into subsequent dry periods (Phinney 2019). It should be noted, however, that greater reproductive investment is also associated with higher STM (Gauslaa 2026).

Thallus thickness generally increases with growth and size (Dahlman et al. 2002; Chinnery and Ellis 2023), suggesting size-related acclimatization to hydration sources (Merinero et al. 2014). In the present study, however, no significant relationship was found between initial transplant size and change in STM ($P=0.727$). Within the limited size range of transplants, the relationship between area and STM (Fig. 4) was weak in both the shade-adapted population at the start ($R^2_{adj}=7.8\%$; $P<0.001$) and in the light-acclimated transplants at the end of experiment ($R^2_{adj}=2.4\%$; $P=0.005$). Even at a given transplant size, STM increased over the course of the study (Fig. 4). Because STM is a functional trait involved in optimization of hydration (Gauslaa 2014; Canali et al. 2025), the vertical distance between the two

Fig. 4 The relationship between specific thallus mass and hydrated thallus area in *Lobaria pulmonaria* transplants sampled from various forest types and climates. The uppermost regression line with downward triangles represents thalli collected from southeastern Norwegian broad-leaved forests (annual rainfall: 1136 mm; Gauslaa et al. 2006). The two lowermost lines show data for transplants from spruce branches in less rainfall-influenced inland forests of southern British Columbia at start (solid black line, filled circles) and end (hatched line, open circles) (Gauslaa and Goward 2012; Bidussi et al. 2013b)

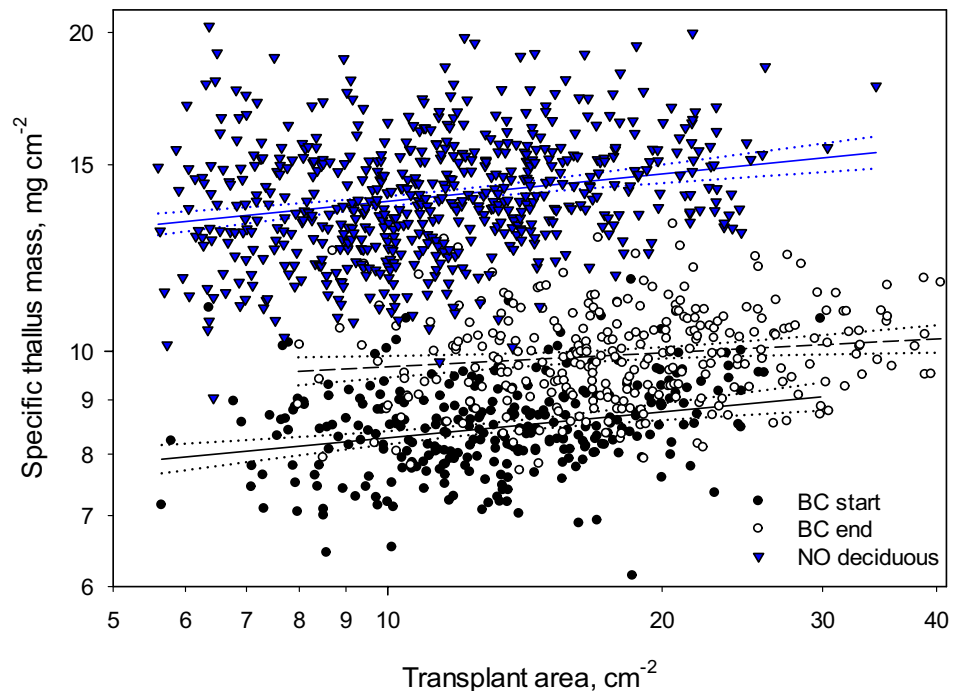


Table 3 Best subset multiple regression model percent increase in specific thallus mass (STM) in *Lobaria pulmonaria* run as linear mixed model after adding the cluster variable: “tree ID” ($R^2_{\text{marginal}}=36.1\%$; $R^2_{\text{conditional}}=39.0\%$)

Term	Coeff ± SE	t-value	P-value
Intercept	17.09 ± 0.44	38.42	<0.001
Direct site factor	-26.6 ± 4.3	-6.18	<0.001
Phosphorus addition (0 or 1)	5.05 ± 0.80	6.33	<0.001
STM _{Start} , mg cm ⁻²	-2.84 ± 0.48	-5.91	<0.001
Elevation, m	0.0076 ± 0.0016	4.91	<0.001
Bark pH	2.52 ± 0.74	3.41	<0.001
Laminal soralia _{End} , no cm ⁻²	0.88 ± 0.45	1.96	0.051

The continuous variables percent increase in STM, elevation, bark pH, specific thallus mass at start (STM_{Start}), direct site factor (DSF), as well as the categorical variable P-addition, were retrieved from Gauslaa and Goward (2012); Bidussi et al. (2013b), while laminal soralia_{End} (which was the soralia variable with the best predictive power for percent change in STM) was from Gauslaa (2026). Log-transformations of variables resulted in weaker models

lower regression lines in Fig. 4—compensating for effects due to thallus size—likely reflects acclimatization from the shaded source site to the more open transplantation sites. For comparison, similarly sized transplants from deciduous forest in southern Norway, where rainfall is greater and dewfall is less important, had the thickest thalli (Fig. 4). Because those transplants likely differed genetically from the BC material, their higher STM may reflect both adaptation and acclimatization.

Most transplants increased in thickness, with STM rising on average by $17.5 \pm 0.5\%$ (range = -2.5–44.5%) from an initial mean of 8.5 ± 0.5 mg cm⁻². This thickening was

statistically significant, as indicated by non-overlapping 95% confidence intervals for STM at start and end of experiment (Fig. 4). Comparable studies in Scandinavian spruce-dominated rainforest reported smaller increases over shorter periods; $6.6 \pm 0.3\%$ after 110 summer days (Gauslaa et al. 2007) and $2.7 \pm 0.7\%$ after 27 days in open sites (Kann et al. 2025). These comparisons suggest that thickening begins rapidly after transplantation but is not completed even after 14 months.

Change in STM was positively correlated with RGR, supporting the idea that thickening contributes to lichen biomass growth (% increase in STM = $10.74 + 6.06 \cdot \text{RGR}$; $R^2_{\text{adj}}=5.30\%$; $P<0.001$), and correlated negatively with RT_AGR (% increase in STM = $22.64 - 65.5 \cdot \text{RT}_{\text{A}}\text{GR}$; $R^2_{\text{adj}}=6.27\%$; $P<0.001$). Additional explanatory variables greatly improved model performance ($R^2_{\text{marginal}}=36.1\%$; $R^2_{\text{conditional}}=39.0\%$; Table 3), although explanatory power remained lower than for the RT_AGR model. Direct site factor (DSF) was strongly and negatively related to STM (Table 3), whereas ISF was positively related to RT_AGR (Table 2). High initial STM reduced the degree of further thickening, while higher bark pH was associated with increases in both STM and RT_AGR. Higher elevation and greater reproductive investment were linked to increased STM (Table 3) but reduced RT_AGR. Notably, the initial P-fertilization significantly increased thickening (Table 3) without affecting biomass or area growth, a finding discussed separately below.

Increased thallus thickness, and consequently greater water storage, is generally interpreted as an acclimatization to elevated evaporative demands and therefore often varies

strongly along environmental gradients. For example, STM is nearly twice as high in *Hypogymnia physodes* on sun-exposed tree trunks and cliffs as on shaded forest trunks (Solhaug et al. 2009), and STM increases with canopy openness in *Lobaria* species (Gauslaa and Coxson 2011). Thallus thickening is typically triggered by transplantation to more light-exposed habitats (Gauslaa 2014; Merinero et al. 2015; Longinotti et al. 2017; Chinnery and Ellis 2023; Di Nuzzo et al. 2026). In the present study, transplants showed slightly increased browning at the end of the experiment (Fig. 1), indicative of melanin formation as a response to higher UV-B levels (Gauslaa and Solhaug 2001; Solhaug et al. 2003). This change is consistent with higher light exposure at transplant sites relative to the source habitat. Seasonal studies have shown that STM reaches the highest values during the driest part of the year in both *L. pulmonaria* (Gauslaa et al. 2013; Di Nuzzo et al. 2026) and *Xanthoria aureola* (Gauslaa and McEvoy 2005), likely because low turgor pressure in fungal hyphae then prevents area expansion while still allowing net photosynthetic gain (Larsson et al. 2012). STM also increases with temperature, as demonstrated in a 14-days growth chamber experiment (Bidussi et al. 2013a), probably because higher temperature increases evaporative demands.

The present study also revealed patterns that contrast with general expectations. A decline in STM with increasing DSF (Fig. 5; $P < 0.001$) was an unexpected finding. This seemingly contradictory response may reflect spatial variations in nocturnal dew formation. On clear nights, cold air from the surrounding mountains tends to accumulate in

topographic depressions such as the valley floor, and stronger nocturnal cooling may therefore generate sufficient morning dew to hydrate lichens at sites with high DSF, where sparse canopy cover provides little protection against long-wave reradiation (Y. Gauslaa, pers. obs.). Hydration to full turgor promotes thallus area growth during the night and early morning (Bidussi et al. 2013a), whereas rapid sun-driven desiccation later in the day shortens the photosynthetically active period and thereby limits further thickening (Larsson et al. 2012; Bidussi et al. 2013a). STM may increase most rapidly when hydration supports prolonged net photosynthesis below the threshold needed for full turgor.

Finally, STM increased with elevation along the steep forested mountainside studied here (Table 3). Higher elevations in the study area are characterized by cooler temperatures and greater rainfall (Bidussi et al. 2013b), whereas stronger nocturnal cooling at lower elevations frequently promotes the formation of heavy dew. Nocturnal dew and humid air tend to favor thallus area extension, resulting in thinner thalli (Bidussi et al. 2013a; Gauslaa 2014), while lichens in habitats where rain is the main hydration source may invest more in thallus thickness to improve water storage and compensate for less frequent hydration events. An elevational increase in STM has also been reported for *Nephroma arcticum* (McEvoy et al. 2007). In that case, however, the gradient extended above the alpine timberline, suggesting that the increase in STM may instead have reflected greater light exposure and correspondingly shorter periods of full-turgor hydration in alpine environments.

Fig. 5 The relationships between percent change in specific thallus mass (STM) and direct site factor for *Lobaria pulmonaria* transplants with (P1) and without (P0) an initial phosphorus addition. For transplants without initial phosphorus addition: percent change in STM = $19.770 - 23.589 \times \text{DSF}$ ($R^2_{\text{adj}} = 8.6\%$; $P < 0.001$). For transplants with phosphorus addition: percent change in STM = $27.131 - 31.155 \times \text{DSF}$ ($R^2_{\text{adj}} = 13.1\%$; $P < 0.001$). Dotted lines indicate 95% confidence intervals

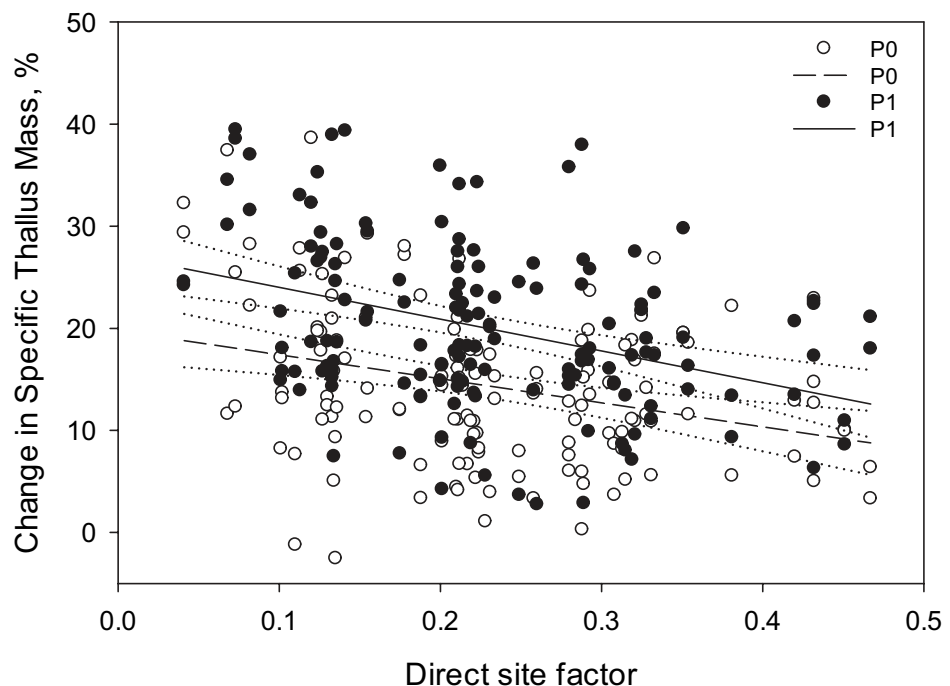


Table 4 Responses of specific thallus mass (STM), internal water holding capacity at end (WHC_{internal}) and various growth parameters such as change in STM, relative growth rate (RGR), and relative thallus area growth rate (RT_AGR) to an initial addition of phosphorus before start of transplantation

	-Phosphorus	+Phosphorus	t-value	P-level
STM _{Start} , mg cm ⁻²	8.56±0.07	8.47±0.07	0.99	0.325
STM _{End} , mg cm ⁻²	9.79±0.08	10.17±0.09	-3.15	0.002
Change in STM, %	14.7±0.7	20.3±0.7	-5.89	<0.001
WHC _{internal} , mg H ₂ O cm ⁻²	12.4±0.1	12.8±0.2	-2.37	0.018
RGR, mg g ⁻¹ d ⁻¹	1.10±0.03	1.15±0.03	-1.30	0.193
RT _A GR, mm ² cm ⁻² d ⁻¹	0.080±0.003	0.073±0.003	1.92	0.056

Means±SE are given. The +P treatment included 5 min submersion of thalli in 500 mg L⁻¹ K₂PO₄, the -P treatment used 5 min submersion of thalli in distilled water. Data were combined from Gauslaa and Goward (2012); Bidussi et al. (2013b)

Phosphorus-induced increases in thallus thickness and water storage

Phosphorus addition was a strong positive predictor for the percentage increase in thallus thickness ($P < 0.001$; Table 3). Its effect was most pronounced in transplants at shaded sites and diminished in more open stands with high DSF, as indicated by overlapping confidence intervals at the highest DSF values (≥ 0.38 ; Fig. 5). Transplants receiving the initial P-fertilization exhibited a significantly greater increase in STM ($20.3 \pm 0.7\%$) than controls ($14.7 \pm 0.7\%$; $P < 0.001$; Table 4). P-treated transplants also had slightly higher final STM ($P = 0.002$) and, correspondingly, higher WHC_{internal} ($P = 0.018$; Table 4). In contrast, RGR was unaffected by the P treatment ($P = 0.193$), and RT_AGR showed a marginal tendency to be lower in P-fertilized transplants ($P = 0.056$; Table 4). Phosphorus addition did not contribute to models of RT_AGR ($P = 0.99$) or RGR ($P = 0.96$).

Within the low to intermediate pH-range typical of tree bark and forest canopy throughfall, increasing pH is often associated with elevated levels of P and base cations, reflecting a shift from oligotrophic to more mesotrophic conditions (Penn and Camberato 2019; Gauslaa et al. 2020). Although both higher bark pH and P addition were associated with enhanced STM (Table 3), their low variance inflation factors (1.04 and 1.13) indicate that these factors were largely independent. This lack of confounding may reflect the fact that oligotrophic conditions—characterized by low bark and soil pH, as well as less favorable throughfall chemistry (Gauslaa et al. 2021)—cannot sustain dense forests, resulting in greater stand openness and higher evaporating demand. Thus, the positive effect of pH on STM (Table 3) is not necessarily attributable to increased P availability, but rather to broader habitat and chemical conditions associated with site fertility and structure.

Lichen are known to rapidly take up P from solutions (Smith 1960; Hyvärinen and Crittenden 1998), and the initial P treatment likely resulted in substantial uptake, potentially lowering the lichen N/P ratio. To compensate for this imbalance (Crittenden et al. 2023; Crittenden and

Thornton 2024), *L. pulmonaria* may have increased investment in cephalodia or enlarged existing ones, possibly at the expense of green algal thallus tissues. Although increased cephalodia in P-treated transplants could have contributed to greater STM, cephalodia occupy only a small portion of total thallus volume (Pentecost 2021; Gerasimova et al. 2021), despite their large size (Millbank and Kershaw 1970). Previous studies have reported mixed effects of P-fertilization. For example P addition did not influence STM or cephalodia mass in *Peltigera aphthosa* (Palmqvist et al. 2017), likely because this ground-dwelling lichen genus shares hydration with thick bryophyte carpets and therefore does not hydrologically benefit from increased thallus thickness in the same way as epiphytic lichens. Another possible response to an N/P imbalance is a shift in the photobiont/mycobiont ratio. Notably, P-treatment in *Cladonia stellaris* increased fungal hyphal growth relative to algal cell numbers (Makkonen et al. 2007), and a similar effect may have occurred in *L. pulmonaria* transplants examined here.

At a broader scale, P-fertilization has been associated with increased abundance of cyanolichens in a Hawaiian montane forest (Benner and Vitousek 2007) and with higher N fixation in *Pseudocyphellaria crocata* (Benner et al. 2007). In one study, an initial P dose doubled biomass growth in *L. pulmonaria* transplants (McCune and Caldwell 2009), whereas in another it had no effect (Marks et al. 2015), highlighting the importance of background P availability and local environmental conditions. The bedrock in the present study area consists of young, P-rich basalts (Hickson 1986; Porder and Ramachandran 2012), which may explain why added P did not stimulate further growth in area or biomass.

Overall, these findings indicate that P enrichment can enhance thallus thickening and water storage capacity in *L. pulmonaria*, especially in shaded habitats, but does not necessarily promote area or biomass expansion. P-induced changes in STM may introduce additional noise into models relating STM to hydration sources.

Conclusion

Rather than following a single growth strategy, *L. pulmonaria* appears to balance area expansion, thallus thickening, and reproduction in ways that depend strongly on local environmental conditions. This helps explain both the marked variability in growth among transplants and the species' close association with special forest habitats. The results suggest that successful persistence of *L. pulmonaria* depends not only on suitable substrate and light conditions, but also on hydration regimes and nutrient context at fine spatial scales. From a conservation perspective, this underlines the importance of maintaining heterogeneous, structurally intact forest habitats that provide the microclimate and chemical conditions required for sustained growth.

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Author contributions YG conceived, designed, and executed this study and wrote the manuscript. No other person is entitled to authorship.

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Declarations

Competing interests The author declares no relevant financial or non-financial interests to disclose.

Consent for publication Not applicable.

Consent to participate Not applicable.

Ethical approval For this type of study formal consent is not required.

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